Physics-Informed LLM Latency Prediction:
Critical Analysis of the Analytical Roofline Model vs. Direct Operational Modeling

Pre-Execution Scheduling & The Token Count Prerequisite

In production LLM serving, latency predictors must operate **strictly pre-execution** using launch configurations: cluster parallelism (`tp`, `pp`, `dp`, `num_gpus`) and request budgets (`batch_size`, `output_tokens`). Post-execution telemetry is strictly excluded.

The Token Prerequisite (Imai et al., Section A.4): The analytical Roofline Model (RLM) calculates hardware ceilings from input and output tokens:

$$T_{\text{roofline}} = T_{\text{prefill}}(\text{input_tokens}) + \text{output_tokens} \times T_{\text{decode_step}}(\text{batch_size}) \tag{1}$$

While prompt length is known pre-execution, output length is dynamic. As acknowledged by Imai et al. (Appendix A.4): *"In practical application, the generation length is not predetermined... To accurately model inference latency in real world scenarios, we must consider predicting the generation length based on the input prompt."*

Methodological Analysis: Single-Model Benchmark Artifact vs. Cross-Model Reality:

- **The Single-Model Confound (Why Trees Treat RLM as Tokens)**: In a fixed-model benchmark (Qwen3-14B) evaluated over discrete prompt lengths (512 vs. 2048), model parameters ($N_{\text{params}}, d_{\text{model}}, L$) and hardware ceilings are constant. Consequently, the analytical roofline T_{roofline} is a strictly monotonic mathematical transform of `input_tokens`. Non-linear tree ensembles (RF/GB) split either feature identically, yielding an identical $R^2 = 0.9997$ (MSE 91.2 vs. 87.8). Within this single-model grid, RLM adds no distinct predictive signal beyond a binary prompt indicator.
- **Why the Roofline Model Is Essential Across Models**: In production serving across heterogeneous model families (e.g., 1.3B to 27B), a raw scalar `input_tokens` carries **zero architectural context**—512 tokens induces orders-of-magnitude different compute/memory load on a 1.3B vs. a 27B model. The analytical Roofline Model bridges this gap by grounding predictions in model FLOPs, tensor dimensions, and hardware bandwidth.
- **Linear Inductive Prior**: For linear regressors that cannot learn non-linear decision boundaries, RLM provides a meaningful physical inductive bias, reducing MSE by 45% (69038 → 37921, $R^2 = 0.7879 \rightarrow 0.8844$) compared to raw token counts.

Table 1: Qwen3-14B: Independent Feature Set Ablation on Standard 3-Fold Cross-Validation (R^2 / MSE).
Base features: `[tp, pp, dp, num_gpus, batch_size, output_tokens]`. Columns 3 and 4 add either `input_tokens` OR `Unscaled RLM` independently to the Base set to evaluate their standalone predictive impact.

Model / Estimator	Base (No Prompt Tok)	Base + input_tokens	Base + Unscaled RLM	Key Takeaway
Linear Regression (LR)	0.3005 / 224574	0.7879 / 69038	**0.8844 / 37921**	RLM aids linear fit
Random Forest (RF)	0.2553 / 236926	**0.9997 / 91.2**	**0.9997 / 87.8**	**Identical tree splits**
Gradient Boosting (GB)	0.2494 / 238559	**0.9997 / 113.8**	0.9995 / 163.7	**Identical tree splits**

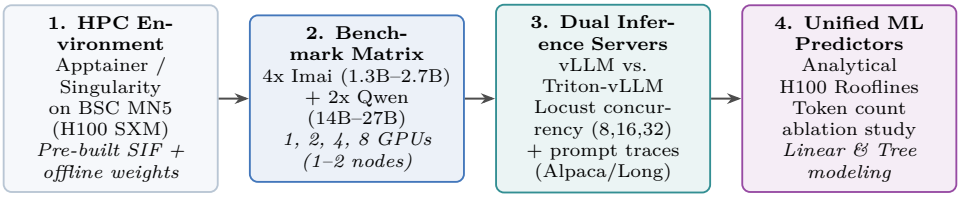
Key Conclusions for Deployment & Next Steps

- **Single-Model Scope**: For isolated single-model deployments where prompt length is known pre-launch, empirical tree models achieve maximum accuracy using raw prompt tokens without analytical profiling.
- **Cross-Model Serving**: The Roofline Model's primary value lies in cross-model and cross-hardware serving matrices, where physical parameter scaling (O(FLOPs), tensor sizes) is mandatory.
- **Value for Linear Estimators**: RLM provides an effective inductive physics prior for linear estimators ($R^2 = 0.7879 \rightarrow 0.8844$, reducing MSE by 45%).
- **Experimental Next Steps**: Validate across continuous sequence lengths (dynamic RAG / chatbot traces) rather than fixed 2-point grids, and test multi-node scaling across 32B/70B architectures.

Multi-GPU Benchmarking & Replication Architecture: BSC MareNostrum 5 (MN5)

Execution Overview: To empirically replicate and scale the Roofline-driven latency prediction methodology of Imai et al., benchmarks are conducted on the **Barcelona Supercomputing Center (BSC) MareNostrum 5 Accelerated Partition** (NVIDIA H100 SXM, 4 GPUs/node). The benchmark matrix combines the 4 original lightweight models (1.3B–2.7B) with 2 production-scale models (14B–27B) across both **vLLM** and **Triton Inference Server**, scaling from single-GPU up to 8 GPUs (2 nodes) under tensor, pipeline, and data parallelism.

1. Experimental Pipeline & Execution Workflow



2. Harmonized Dataset Schema: Integrating Imai et al. with Existing Benchmarks

To enable unified cross-model and cross-engine modeling, we merge Imai et al.'s load-testing parameters with our cluster parallelism features. Columns newly introduced for the BSC HPC campaign are highlighted in color:

Feature Group	Column Name	Data Type / Domain	Methodological Purpose & Description
Model Topology	`model_name, num_params`	Categorical / Integer	Architecture identification across scales (1.3B to 27B parameters).
	`tp_degree, pp_degree, dp_degree`	Integer ($\{1, 2, 4\}$)	Parallelism tensor/pipeline/data degrees.
	num_nodes, num_gpus	Integer ($1, 2$ nodes; $1 \ldots 8$ GPUs)	**NEW (HPC):** Multi-node scaling over 800 Gb/s NDR200 InfiniBand fabric.
Serving Context	**server_engine**	`vLLM` vs. `Triton-vLLM`	**NEW (Engine):** Tests runtime overhead gap identified by Imai et al.
	load_regime	`locust` vs. `trace_batch`	**NEW (Load):** Disentangles concurrent client requests from static batch sizes.
	concurrent_users	Integer $(8, 16, 32)$ / N/A	**NEW (Imai):** Locust concurrency parameter used in Imai et al. Table 1.
	`input_tokens, output_tokens`	Integer $(32 \ldots 2048)$	Prompt token count and max generation budget (pre-execution features).
Physical Ceilings	`roofline_analytical_latency`	Float (Seconds)	Unscaled theoretical single-GPU execution ceiling via LLM-Viewer.
Ground Truth	`duration, throughput, power`	Float (ms, tok/s, W)	Measured end-to-end latency (regression target), throughput, and energy.

3. HPC Compute Allocation Budget (BSC MareNostrum 5 Accelerated Partition)

MareNostrum 5 provides 4x NVIDIA H100 SXM (64GB) per node. Testing 8 GPUs utilizes 2 nodes over NDR200 InfiniBand. Below is the compute budget breakdown to collect all required benchmark data:

Experimental Campaign	Evaluated Models	Hardware Setup	Total Configurations	Wall Time	Compute Budget
Phase 1A: Imai Replication	4 models (1.3B–2.7B)	1 GPU (SXM), vLLM + Triton	9 configs × 3 reps × 2 servers	≈ 18 hours	**18 GPU-h**
Phase 1B: Multi-GPU Scaling	4 models (1.3B–2.7B)	TP$\{2, 4\}$, DP$\{2, 4\}$ (1 node)	4 configs × 6 workloads × 3 reps	≈ 8 hours	**28 GPU-h**
Phase 2: Triton Qwen Runs	Qwen3-14B / 27B	1, 2, 4, 8 GPUs (1–2 nodes)	Matches existing 450 vLLM runs	≈ 15 hours	**60 GPU-h**
Phase 3: Roofline Ceilings	All 6 target models	Offline CPU analytical model	54 discrete hardware ceilings	< 1 hour	**0 GPU-h (CPU)**
Total HPC Allocation	**6 Architectures**	**1 to 8 H100 GPUs**	∼ 1,180 **executed trials**	≈ 41 **hours**	≈ 106 **GPU-h**